

MOHD ANAS

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EDUCATION

B.Tech, Information Technology

IIIT Vadodara, Gujarat, India

Expected Graduation: Aug 2027

7.46 / 10.0 GPA

Coursework: Backend System, Operating Systems, OOP, Data Structures & Algorithms, Machine Learning, Deep Learning

PROFESSIONAL EXPERIENCE

Aarogya Kaya | SDE Intern | Remote

July 2025 - Jan 2026

- Architected Go Backend services for Shiprocket and Instagram Graph API integrations, implementing **CQRS architecture** and a **three-tier caching strategy (in-memory, Redis, and database)**, reducing average read latency from 420ms to 38ms (91%) and maintaining p95 response times below 250ms.
- Refactored 15+ Backend modules using **SOLID principles** and **Design Patterns** (such as **Dependency Injection**), reducing code duplication by 30% and improving feature delivery speed.
- Engineered fault-tolerant asynchronous synchronization pipelines with retry mechanisms and **idempotent processing**, supporting **500+ daily events** while maintaining **99.9%** data consistency.

Open Source Contributor | Remote

July 2025 - Present

- Contributed a merged Pull Request to **Next.js (Vercel)**, migrating the patch-next utility from CommonJS to TypeScript ESM and **reducing contributor setup overhead** by eliminating repeated dependency reinstalls.
- Contributed **22 merged Pull Requests** and opened **12 Issues** across open-source projects, fixing bugs, implementing features, and improving developer tooling.

PROJECTS

Second Brain | Agentic Career Path Simulator | Python, LLMs, Redis Queue, React Flow

Live | GitHub

- Architected an asynchronous LLM job processing pipeline using **Redis Queue** and background workers, decoupling multi-step agent execution loops and cutting API latency by **65%** (15s to 5.2s).
- Designed a dynamic context-retrieval and schema-enforced validation engine, ensuring deterministic structured JSON outputs from nondeterministic model runs with a **98.2%** job completion success rate.
- Built a real-time decision-graph visualizer supporting **100+ simulated path nodes** at **60 FPS** using **React Flow** and **Zustand** to render complex LLM tradeoff analysis.

a3.store | Distributed Key-value Store | Python, EKS, AWS, Kubernetes, Prometheus

Live | GitHub

- Developed a **leaderless** Distributed key-value store implementing replication, **failure detection**, synchronization, gossip-based membership, and anti-entropy repair across a 3-node cluster.
- Deployed the system on **AWS EKS** using Kubernetes StatefulSets and persistent volumes, managing rolling upgrades, pod restarts, and **self-healing behavior**.
- Architected internal modules using **SOLID principles** and clean interfaces, decoupling storage engines from network transport to enable pluggable consensus mechanisms and isolated unit testing.

TECHNICAL SKILLS

Languages: Python, Go, C++, TypeScript, Javascript, SQL, React, Node.js, gRPC

Tools & Technologies: Git, GitHub, Docker, **Kubernetes**, **AWS**, EC2, EKS, IAM, VPC, Terraform, **Linux**, Kafka, Jenkins, Redis, SQLite, Prometheus, Grafana, Loki

AI & Agentic Systems: Multi-Agent Orchestration, Model Context Protocol (MCP), RAG, Vector Search, LLM Tool Calling, Prompt Caching, Prompt Engineering, Context Engineering

ACHIEVEMENT

- Achieved Codeforces **Pupil (1292)**.
- **Solved 500+** DSA problems across **Codeforces**, **LeetCode**, **CodeChef**, and **GeeksforGeeks**.

LEADERSHIP & ACTIVITIES

- Organized an intra-college hackathon and mentored **25+ teams** as a **Technical Committee Core Member**.
- Organized **5+** events and workshops, driving student engagement in **competitive programming** as **Coding Club Member**.